European Journal of Control xxx (xxxx) xxx

Contents lists available at ScienceDirect

European Journal of Control

journal homepage: www.sciencedirect.com/journal/european-journal-of-control

A family of degenerate robust Kalman filters ⋆

Mattia Zorzi*, Shenglun Yi

Department of Information Engineering, University of Padova, Via Gradenigo 6/B, 35131, Padova, 35131, Italy

ARTICLE INFO

Recommended by T Parisini

Keywords:
Robust filtering; mini-max games; Riccati equations; divergence families.

ABSTRACT

A well-established way to address model uncertainty in state estimation problems is through minimax formulations, where the optimal filter is obtained against the least favorable model within an ambiguity set. The latter is defined by placing an upper bound on a divergence between the actual and nominal state-space models. In this work, we extend the robust Kalman filtering framework to the case in which the ambiguity set is defined using the τ-divergence family, and the Riccati recursion evolves on the cone of positive semi-definite matrices. The resulting family of robust Kalman filters, parameterized by τ, includes as a special case a degenerate robust Kalman filter previously introduced in the literature. Moreover, it generalizes the existing class of robust Kalman filters based on the τ-divergence, whose Riccati recursion is restricted to the cone of positive definite matrices. We derive the corresponding least favorable model, study the convergence properties of the Riccati iteration under constant model parameters, and present numerical experiments showing that the proposed estimator provides a flexible framework where an appropriate tuning of the parameter τ can lead to improved transient performance.

1. Introduction

A well-established approach to address model uncertainty in state estimation problems is to determine the optimal filter by means of a minimax game, Başar and Bernhard (2008), Hansen and Sargent (2008), Kim et al. (2020), Nguyen et al. (2023). In this framework, one player aims to identify the least favorable model within a specified ambiguity set, while the other player seeks the optimal estimator tailored to this worst case scenario. A key aspect of this approach lies in the design of the ambiguity set, in which the modeling mismatch is measured through a divergence measure. More precisely, the ambiguity set is a ball formed by placing an upper bound on the divergence between the nominal and the actual model. The most popular one is the Kullback–Leibler (KL) divergence (Levy & Nikoukhah, 2004, 2013; Yoon et al., 2004), which has subsequently been extended to the τ-divergence family (Zorzi, 2016), where the parameter τ shapes the corresponding ambiguity set used to model uncertainty (Zorzi, 2017b). The appealing feature of this approach is that the optimal estimator exhibits a structure similar to the Kalman filter, i.e. simple, with the key difference being a distortion in the Riccati recursion.

In many application scenarios the covariance matrix of the Riccati recursion may become singular (i.e. low-rank), see for instance, Bocquet et al. (2017), Bonnabel and Sepulchre (2013), Song et al. (2014), Zorzi and Yi (2024). In this case, the resulting filter is commonly referred to as a degenerate Kalman filter. While this poses no issue for the standard Kalman filter, the robust filter based on the τ-divergence fails because the Riccati recursion is defined on the cone of positive definite matrices.

In this paper, we derive an extension of the robust Kalman filtering approach using the τ-divergence proposed in Zorzi (2016) to the degenerate case, which naturally defines a family of robust Kalman filters parameterized by τ. The proposed approach is very flexible; indeed, (i) unlike (Zorzi, 2016), the Riccati recursion is not restricted to the cone of positive definite matrices but is defined on the cone of positive semi-definite matrices, and (ii) it includes as a special case the estimator considered in Yi and Zorzi (2022) (for $\tau = 0$). The robust estimator is characterized by a Riccati iteration defined on the cone of positive semi-definite matrices. We further derive the least favorable model and provide a convergence analysis of the Riccati iteration under the assumption that the nominal model has constant parameters. Finally, we present numerical experiments indicating that the proposed filter offers increased flexibility, with appropriate choices of τ yielding better transient behavior than Yi and Zorzi (2022).

The outline of the paper is as follows. In Section 2 we derive the static estimation problem which will be used in Section 3 to derive the degenerate robust estimation problem. In Sections 4 and 5 we derive the least favorable model and analyze the convergence of the Riccati iteration, respectively. In Section 6 the numerical experiments are presented and in Section 7 we draw the conclusions.

⋆ This work was partially supported by the Italian Ministry of University and Research under the PRIN project “Extracting essential information and dynamics from complex networks”, grant no. 2022MBC2EZ.

* Corresponding author.
E-mail address: zorzimat@dei.unipd.it (M. Zorzi).

https://doi.org/10.1016/j.ejcon.2026.101562
Received 4 June 2026; Accepted 18 July 2026
Available online 23 July 2026

0947-3580/© 2026 The Authors. Published by Elsevier Ltd on behalf of European Control Association. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Notation: $\mathcal{Q}_+^n$ ($\overline{\mathcal{Q}_+^n}$) denotes the cone of n-dimensional symmetric matrices which are positive (semi)-definite. Given $X \in \mathbb{R}^{n\times n}$, $X^\dagger$ and $\exp(X)$ denote the pseudo inverse and the matrix exponential, respectively; $\mathrm{Im}(X)$ and $\|X\|_Q$, with $Q \in \mathcal{Q}_+^n$, denote the image and the weighted norm of X. Given $X \in \overline{\mathcal{Q}_+^n}$: X^τ, with $\tau > 0$, denotes the matrix power; $\log^\dagger(X)$ denotes the pseudo matrix logarithm, i.e. the matrix obtained by replacing each positive eigenvalue of X with its logarithm and leaving the zero eigenvalues unchanged; $\sigma_{\max}(X)$ denotes the maximum eigenvalue of X; $X^{\dagger\tau}$, with $\tau > 0$, denote the pseudo inverse of X^τ. Finally, $f \sim \mathcal{N}(\mu, X)$ means that f is the probability density of a Gaussian random vector with mean μ and covariance matrix X.

2. Static estimation problem

Consider the task of estimating a random vector $x \in \mathbb{R}^n$ based on an observation vector $y \in \mathbb{R}^p$. The nominal probability density function of $z := [\,x^\top\ y^\top\,]^\top$ is $f(z) \sim \mathcal{N}(m_z, K_z)$ with $m_z \in \mathbb{R}^{n+p}$ and $K_z \in \overline{\mathcal{Q}_+^{n+p}}$ such that

$$m_z = \begin{bmatrix} m_x \\ m_y \end{bmatrix}, \quad K_z = \begin{bmatrix} K_x & K_{xy} \\ K_{yx} & K_y \end{bmatrix}. \tag{1}$$

We assume that K_z is possibly a singular matrix and such that $\mathrm{rank}(K_z) = r + p$ with $r \le n$ and $K_y \in \mathcal{Q}_+^p$. Therefore, the support of $f(z)$ is the $r+p$-dimensional affine subspace $\mathcal{A} = \{m_z + v, \quad v \in \mathrm{Im}(K_z)\}$. We assume that the actual probability $\tilde f(z) \sim \mathcal{N}(\tilde m_z, \tilde K_z)$ is unknown and has the same support $\mathcal{A}$ of $f(z)$. Accordingly, $\mathrm{Im}(K_z) = \mathrm{Im}(\tilde K_z)$, $\Delta m_z \in \mathrm{Im}(K_z)$ where $\Delta m_z = \tilde m_z - m_z$. Since f and $\tilde f$ are completely described by their mean and covariance matrix, we measure the discrepancy between them using the τ-divergence, Zorzi (2017b). The latter, however, is only defined for non-degenerate Gaussian densities, i.e. it is not able to measure "deterministic" deviations. Therefore, we adapt such notion as follows

$$\mathcal{D}_\tau(\tilde f, f) = \begin{cases} \|\Delta m_z\|^2_{K_Z^\dagger} + \mathrm{tr}\left(-\log^\dagger(\tilde K_z K_z^\dagger) + \tilde K_z K_z^\dagger\right) - (p+r), & \tau = 0 \\ \|\Delta m_z\|^2_{\frac{1}{1-\tau}K_Z^\dagger} + \mathrm{tr}\left(-\frac{1}{\tau(1-\tau)}(L_z^\dagger \tilde K_z (L_z^\dagger)^\top)^\tau \right. & \\ \quad \left. + \frac{1}{1-\tau}\tilde K_z K_z^\dagger\right) + \frac{p+r}{\tau}, & 0 < \tau < 1 \\ \delta_\infty(\Delta m_z) + \mathrm{tr}\left(L_z^\dagger \tilde K_z (L_z^\dagger)^\top \log^\dagger(L_z^\dagger \tilde K_z (L_z^\dagger)^\top)\right. & \\ \quad \left. - \tilde K_z K_z^\dagger\right) + p + r, & \tau = 1 \end{cases} \tag{2}$$

where L_z is a square root matrix of K_z such that $K_z = L_z L_z^\top$, $\Delta m_z = \tilde m_z - m_z$,

$$\delta_\infty(v) := \begin{cases} 0 & \text{if } v = 0 \\ \infty & \text{otherwise.} \end{cases}$$

It is straightforward to see that (2) coincides with the standard τ-divergence in the case $\tilde f$ and f are non-degenerate. Notice that, the definition in (2), as well as the next derivations, do not depend on the particular choice of L_z.

Lemma 1. *Let $f(z) \sim \mathcal{N}(m_z, K_z)$ and $\tilde f(z) \sim \mathcal{N}(\tilde m_z, \tilde K_z)$ be Gaussian and possibly degenerate probability density functions with the same $r+p$-dimensional support $\mathcal{A}$. Let*

$$\mathcal{U} = \{\tilde m_z \in \mathbb{R}^{n+p} \text{ s.t. } \tilde m_z - m_z \in \mathrm{Im}(K_z)\}$$
$$\mathcal{V} = \{\tilde K_z \in \mathcal{Q}^{n+p} \text{ s.t. } \mathrm{Im}(K_z) = \mathrm{Im}(\tilde K_z)\}.$$

Then, $\mathcal{D}_\tau(\tilde f, f)$ is strictly convex with respect to $\tilde m_z \in \mathcal{U}$ and $\tilde K_z \in \mathcal{V} \cap \overline{\mathcal{Q}_+^{n+p}}$. Moreover, $\mathcal{D}_\tau(\tilde f, f) \ge 0$ and equality holds if and only if $f = \tilde f$.

Proof. Let $U_{\mathfrak r}^\top$ be a matrix whose columns form an orthonormal basis for $\mathrm{Im}(K_z)$. We define the non-degenerate reduced densities, $f_{\mathfrak r} \sim \mathcal{N}(m_z^{\mathfrak r}, K_z^{\mathfrak r})$ and $\tilde f_{\mathfrak r} \sim \mathcal{N}(\tilde m_z^{\mathfrak r}, \tilde K_z^{\mathfrak r})$ where $m_z^{\mathfrak r} = U_{\mathfrak r} m_z$ and $\tilde m_z^{\mathfrak r} = U_{\mathfrak r}\tilde m_z$, while $K_z^{\mathfrak r} = U_{\mathfrak r} K_z U_{\mathfrak r}^\top$ and $\tilde K_z^{\mathfrak r} = U_{\mathfrak r}\tilde K_z U_{\mathfrak r}^\top$ are positive definite. Since $\tilde m_z, m_z \in \mathcal{U}$ and $\tilde K_z, K_z \in \mathcal{V}$, we can obtain f and $\tilde f$ from $\tilde f_{\mathfrak r}$ and $f_{\mathfrak r}$, respectively. More precisely, their mean and covariance matrices are given by $m_z = U_{\mathfrak r}^\top m_z^{\mathfrak r}$, $\tilde m_z = U_{\mathfrak r}^\top \tilde m_z^{\mathfrak r}$, $K_z = U_{\mathfrak r}^\top K_z^{\mathfrak r} U_{\mathfrak r}$ and $\tilde K_z = U_{\mathfrak r}^\top \tilde K_z^{\mathfrak r} U_{\mathfrak r}$. Accordingly, there is a one-to-one correspondence between $(\tilde f, f)$ and $(\tilde f_{\mathfrak r}, f_{\mathfrak r})$. By direct computation, we have $\mathcal{D}_\tau(\tilde f, f) = \mathcal{D}_\tau(\tilde f_{\mathfrak r}, f_{\mathfrak r})$. Thus, the claim follows from the facts that, see Zorzi (2017b): i) $\mathcal{D}_\tau(\tilde f_{\mathfrak r}, f_{\mathfrak r}) \ge 0$ and equality holds if and only if $\tilde f_{\mathfrak r} = f_{\mathfrak r}$; ii) $\mathcal{D}_\tau(\tilde f_{\mathfrak r}, f_{\mathfrak r})$ is strictly convex with respect to $\tilde m_z^{\mathfrak r} \in \mathbb{R}^{r+p}$ and $\tilde K_z^{\mathfrak r} \in \mathcal{Q}_+^{r+p}$. □

We assume that the actual density belongs to the ambiguity set about the nominal density

$$\mathcal{B} = \{\tilde f \sim \mathcal{N}(\tilde m_z, \tilde K_z) \text{ s.t. } \mathcal{D}_\tau(\tilde f, f) \le c\}$$

where $c > 0$ defines the radius of this ball (and is called tolerance), while τ the shape of this ball. Then, we characterize the estimator of x through the following minimax problem:

$$(\hat x, \tilde f^0) = \arg\min_{g\in\mathcal{G}} \max_{\tilde f\in\mathcal{B}} J(\tilde f, g) \tag{3}$$

where

$$J(\tilde f, g) = \frac{1}{2}\int_{\mathcal{A}} \|x - g(y)\|^2 \tilde f(z)\, dz;$$

$\mathcal{G}$ is the set of estimators for which the estimator has bounded error variance with respect to all the Gaussian densities in $\mathcal{B}$.

Theorem 1. *Let f be a Gaussian (possibly degenerate) density with mean and covariance matrix defined as in (1) with $K_y \in \mathcal{Q}_+^p$. Then, the Bayesian estimator*

$$g^0(y) = G_0(y - m_y) + m_x, \tag{4}$$

with $G_0 = K_{xy}K_y^{-1}$, is the minimizer of (3). Moreover, the least favorable Gaussian density $\tilde f^0$ is with mean vector and covariance matrix

$$\tilde m_z^0 =: m_z = \begin{bmatrix} m_x \\ m_y \end{bmatrix}, \quad \tilde K_z^0 := \begin{bmatrix} \tilde K_x & K_{xy} \\ K_{yx} & K_y \end{bmatrix} \tag{5}$$

where $\tilde K_x := \tilde P + K_{xy}K_y^{-1}K_{yx}$. The nominal posterior covariance matrix of x given y is given by

$$P := K_x - K_{xy}K_y^{-1}K_{yx}. \tag{6}$$

while the least favorable one is:

$$\tilde P := \begin{cases} L_P\left(H^\top H - \theta(1-\tau)L_P^\top L_P\right)^{\dagger\frac{1}{1-\tau}} L_P^\top, & 0 \le \tau < 1 \\ L_P \exp\left(\theta L_P^\top L_P\right) L_P^\top, & \tau = 1, \end{cases} \tag{7}$$

where $H^\top \in \mathbb{R}^{n\times r}$ is a matrix whose columns form an orthonormal basis for $\mathrm{Im}(P)$ and $L_P \in \mathbb{R}^{n\times n}$ is such that $P = L_P L_P^\top$. Moreover, $\theta < ((1-\tau)\sigma_{\max}(P))^{-1}$ is the unique solution to the equation $\gamma_\tau(P, \theta) = c$ where

$$\gamma_\tau(P,\theta) := \begin{cases} -\log\det(I_r - \theta P)^\dagger + \mathrm{tr}((I_n - \theta P)^\dagger - I_r), & \tau = 0 \\ \mathrm{tr}\left(\frac{1}{\tau(\tau-1)}\left(H^\top H - \theta(1-\tau)L_P^\top L_P\right)^{\dagger\frac{\tau}{1-\tau}}\right. & \\ \quad \left. + \frac{1}{1-\tau}\left(H^\top H - \theta(1-\tau)L_P^\top L_P\right)^{\dagger\frac{1}{1-\tau}} + \frac{1}{\tau} I_r\right), & 0 < \tau < 1 \\ \mathrm{tr}\left(\exp(\theta L_P^\top L_P)(\theta L_P^\top L_P - H^\top H) + I_r\right), & \tau = 1. \end{cases} \tag{8}$$

Proof. We prove the claim for $0 < \tau < 1$, as the cases $\tau = 0$ and $\tau = 1$ follow similarly. Since $\mathcal{G}$ and $\mathcal{B}$ are convex and compact sets (see Lemma 1), then there exists a saddle point $(\tilde f^0, g^0)$ solution to (3), i.e.

$$J(\tilde f, g^0) \le J(\tilde f^0, g^0) \le J(\tilde f^0, g) \tag{9}$$

for all $\tilde f \in \mathcal{B}$ and $g \in \mathcal{G}$. Notice that, under the assumption the underlying model is described by the Gaussian density $\tilde f^0$, $g^0(y)$ in (4) represents the Bayesian estimator which is the minimum mean square error (MMSE) estimator, Kailath et al. (2000), i.e. it minimizes $J(\tilde f^0, g)$ over $g \in \mathcal{G}$. Thus, the minimizer satisfying the second inequality in (9) coincides with (4). It remains to show that the maximizer $\tilde f^0 \sim \mathcal{N}(\tilde m_z, \tilde K_z)$

satisfying the first inequality in (9) has mean and covariance matrix defined as in (5). First, the minimizer of $J(\tilde f,\cdot)$ is the Bayesian estimator

$$g^\star(y)=\tilde K_{xy}\tilde K_y^{-1}(y-\tilde m_y)+\tilde m_x$$

where

$$\tilde m_z:=\begin{bmatrix}\tilde m_x\\ \tilde m_y\end{bmatrix},\quad \tilde K_z:=\begin{bmatrix}\tilde K_x & \tilde K_{xy}\\ \tilde K_{yx} & \tilde K_y\end{bmatrix}.$$

Moreover, $J(\tilde f,g^\star)=1/2\,\mathrm{tr}(\tilde P)$ where $\tilde P:=\tilde K_x-\tilde K_{xy}\tilde K_y^{-1}\tilde K_{yx}$. Since $\mathrm{Im}(\tilde K_z)=\mathrm{Im}(K_z)$, and in view of (6), it follows that $\mathrm{Im}(\tilde P)=\mathrm{Im}(P)=\mathrm{Im}(H^\top)$. Thus,

$$J(\tilde f,g^\star)=\frac{1}{2}\,\mathrm{tr}(\tilde P H^\top H)=\frac{1}{2}\int\|H(x-g(y))\|^2\tilde f(z)dz.$$

We define the reduced random vector

$$z_{\mathfrak r}:=\begin{bmatrix}x_{\mathfrak r}\\ y\end{bmatrix}=U_{\mathfrak r}z,\quad U_{\mathfrak r}:=\begin{bmatrix}H & 0\\ 0 & I\end{bmatrix}$$

and it is not difficult to see that $z=U_{\mathfrak r}^\top z_{\mathfrak r}$, i.e. there exists a one-to-one correspondence between z and $z_{\mathfrak r}$. Accordingly, the maximization can be formulated in terms of the actual density of $z_{\mathfrak r}$. More precisely, the nominal and actual densities of $z_{\mathfrak r}$ are, respectively, $f_{\mathfrak r}\sim\mathcal N(m_z^{\mathfrak r},K_z^{\mathfrak r})$ and $\tilde f_{\mathfrak r}\sim\mathcal N(\tilde m_z^{\mathfrak r},\tilde K_z^{\mathfrak r})$ where $m_z^{\mathfrak r}=U_{\mathfrak r}m_z$, $\tilde m_z^{\mathfrak r}=U_{\mathfrak r}\tilde m_z$, $K_z^{\mathfrak r}=U_{\mathfrak r}K_zU_{\mathfrak r}^\top$, $\tilde K_z^{\mathfrak r}=U_{\mathfrak r}\tilde K_zU_{\mathfrak r}^\top$. Accordingly, the equivalent problem consists in maximizing

$$J(\tilde f_{\mathfrak r},g_{\mathfrak r}^0)=\int\|x_{\mathfrak r}-g_{\mathfrak r}^0(y)\|^2\tilde f_{\mathfrak r}dz_{\mathfrak r}$$

where $\tilde f_{\mathfrak r}\in\mathcal B_{\mathfrak r}:=\{\tilde f_{\mathfrak r}\sim\mathcal N(\tilde m_z^{\mathfrak r},\tilde K_z^{\mathfrak r})\text{ s.t. }D_\tau(\tilde f_{\mathfrak r},f_{\mathfrak r})\le c\}$, $g_{\mathfrak r}^0:=Hg^0$ and $D_\tau(\tilde f_{\mathfrak r},f_{\mathfrak r})=D_\tau(\tilde f,f)$. Since $\tilde f_{\mathfrak r}$ and $f_{\mathfrak r}$ are non-degenerate densities, i.e. $K_z^{\mathfrak r},\tilde K_z^{\mathfrak r}\in\mathcal Q_+^{r+p}$, by Theorem 3.1 in Zorzi (2017b), we have that the maximizer $\tilde f_{\mathfrak r}^0$ has mean $m_z^{\mathfrak r}$ and covariance matrix

$$\tilde K_z^{\mathfrak r}=\begin{bmatrix}\tilde K_x^{\mathfrak r} & K_{xy}^{\mathfrak r}\\ K_{yx}^{\mathfrak r} & K_y\end{bmatrix}$$

where $\tilde K_x^{\mathfrak r}=\tilde P^{\mathfrak r}+K_{xy}^{\mathfrak r}K_y^{-1}K_{yx}^{\mathfrak r}$ with

$$\tilde P^{\mathfrak r}=L_{P^{\mathfrak r}}\left(I_r-\theta(1-\tau)L_{P^{\mathfrak r}}^\top L_{P^{\mathfrak r}}\right)^{-\frac{1}{1-\tau}}L_{P^{\mathfrak r}}^\top,\tag{10}$$

$P^{\mathfrak r}:=HPH^\top\in\mathcal Q_+^r$, $L_{P^{\mathfrak r}}\in\mathbb R^{r\times r}$ is such that $P^{\mathfrak r}=L_{P^{\mathfrak r}}L_{P^{\mathfrak r}}^\top$ and $\theta<((1-\tau)\sigma_{\max}(P^{\mathfrak r}))^{-1}=((1-\tau)\sigma_{\max}(P))^{-1}$ is the unique solution to

$$\mathbf w(\theta):=\mathrm{tr}\left(\frac{1}{\tau(\tau-1)}\left(I_r-\theta(1-\tau)L_{P^{\mathfrak r}}^\top L_{P^{\mathfrak r}}\right)^{-\frac{\tau}{1-\tau}}+\frac{1}{1-\tau}\left(I_r-\theta(1-\tau)L_{P^{\mathfrak r}}^\top L_{P^{\mathfrak r}}\right)^{-\frac{1}{1-\tau}}+\frac{1}{\tau}I_r\right)=c\tag{11}$$

with r dimension of $P^{\mathfrak r}$. Notice that, the existence of such θ follows from the fact that $\mathbf w(\theta)$ is a continuous function on the open interval $(0,((1-\tau)\sigma_{\max}(P))^{-1})$ such that $\mathbf w(0)=0$ and $\mathbf w(\theta)\to+\infty$ as $\theta\to((1-\tau)\sigma_{\max}(P))^{-1}$. The uniqueness of θ follows from the fact that $\mathbf w(\theta)$ is a strictly increasing function. We conclude that the least favorable density $\tilde f^0(z)$ has mean and covariance matrix as in (5) where

$$\tilde K_x=H^\top\tilde K_x^{\mathfrak r}H=\underbrace{H^\top\tilde P^{\mathfrak r}H}_{=:\tilde P}+\underbrace{H^\top K_{xy}^{\mathfrak r}}_{=K_{xy}}K_y^{-1}K_{yx}^{\mathfrak r}H.$$

Finally, defining $L_P=H^\top L_{P^{\mathfrak r}}H\in\mathbb R^{n\times n}$ we have that $P=L_PL_P^\top$ and by (10), (11) we obtain (7), (8), respectively. □

In the case f is a non-degenerate density in the theorem above we have $H^\top H=I_n$ and pseudo power is substituted by the inverse power, i.e. we recover the robust static estimation problem in Zorzi (2017b).

3. Robust filtering problem

Consider the nominal state space model

$$\begin{aligned}x_{t+1}&=A_tx_t+B_tv_t\\ y_t&=C_tx_t+D_tv_t\end{aligned}\tag{12}$$

where $x_t\in\mathbb R^n$ is the state process, $y_t\in\mathbb R^p$ is the observation process, v_t is white Gaussian noise with unit variance, i.e. $\mathbb E[v_tv_t^\top]=I$, and v_t is independent from the initial state x_0 whose nominal distribution is $f_0(x_0)\sim\mathcal N(\hat x_0,\tilde P_0)$. We assume that $D_tD_t^\top$ is invertible. Thus, without loss of generality, we assume that $\Gamma_t:=[\,B_t^\top\;\;D_t^\top\,]^\top\in\mathbb R^{(n+p)\times(r_t+p)}$ is a full column rank matrix and $v_t\in\mathbb R^{r_t+p}$. If this is not the case, there always exists an equivalent lower-dimensional representation of v_t for which this assumption holds. Such model is characterized by the nominal transition probability density of $z_t:=\begin{bmatrix}x_{t+1}^\top & y_t^\top\end{bmatrix}^\top$ given x_t:

$$\phi_t(z_t|x_t)\sim\mathcal N\left(\begin{bmatrix}A_t\\ C_t\end{bmatrix}x_t,\begin{bmatrix}B_t\\ D_t\end{bmatrix}\begin{bmatrix}B_t^\top & D_t^\top\end{bmatrix}\right).\tag{13}$$

We assume the true model deviates from (12) and we follow the robust approach of Levy and Nikoukhah (2013) to define the state estimator which is aware to uncertainty. Let $\tilde\phi_t(z_t|x_t)$ denote the actual transition density and $\tilde f_t(x_t|Y_{t-1})$ the actual *a priori* density of x_t given $Y_{t-1}:=\{y_s:0\le s\le t-1\}$. We define:

$$\bar f_t(z_t|Y_{t-1})=\int\phi_t(z_t|x_t)\tilde f_t(x_t|Y_{t-1})\,\mathrm dx_t,\tag{14}$$

$$\tilde f_t(z_t|Y_{t-1})=\int\tilde\phi_t(z_t|x_t)\tilde f_t(x_t|Y_{t-1})\,\mathrm dx_t.\tag{15}$$

Here, $\bar f_t(z_t|Y_{t-1})$ acts as a pseudo-nominal density, and the latter is Gaussian under the assumption

$$\tilde f_t(x_t|Y_{t-1})\sim\mathcal N(\hat x_t,\tilde P_t).\tag{16}$$

Assuming that the actual density $\tilde f_t(z_t|Y_{t-1})$ is Gaussian too, the mismatch between $\tilde\phi_t$ and ϕ_t can be measured through the τ-divergence of $\tilde f_t$ and $\bar f_t$. Thus, we assume that the actual density $\tilde\phi_t$ belongs to the ambiguity set $\mathcal B_{t,\tau}=\{\tilde\phi_t:\mathcal D_\tau(\tilde f_t,\bar f_t)\le c_t\}$, where $c_t\ge0$ is the tolerance allowed at time step t. Then, the robust filter is defined by the minimax problem:

$$\min_{g_t\in\mathcal G_t}\max_{\tilde\phi_t\in\mathcal B_{t,\tau}}J_t(\tilde\phi_t,g_t),\tag{17}$$

where

$$J_t(\tilde\phi_t,g_t)=\int\int\|x_{t+1}-g_t(y_t)\|^2\tilde\phi_t(z_t|x_t)\tilde f_t(x_t|Y_{t-1})\,\mathrm dx_t\mathrm dz_t,$$

represents the mean square error of the estimator $g_t(y_t)$ and we require that $\tilde\phi_t(z_t|x_t)$ obeys the normalization condition

$$\int\int\tilde\phi_t(z_t|x_t)\tilde f_t(x_t|Y_{t-1})\mathrm dz_t\mathrm dx_t=1.\tag{18}$$

By (13) and (16), we have

$$\bar f_t(z_t|Y_{t-1})\sim\mathcal N\left(m_{z_t},K_{z_t}\right)\tag{19}$$

with $m_{z_t}=\begin{bmatrix}A_t^\top & C_t^\top\end{bmatrix}^\top\hat x_t$,

$$\begin{aligned}K_{z_t}&=\begin{bmatrix}K_{x_{t+1}} & K_{x_{t+1},y_t}\\ K_{y_t,x_{t+1}} & K_{y_t}\end{bmatrix}\\ &=\begin{bmatrix}A_t\\ C_t\end{bmatrix}\tilde P_t\begin{bmatrix}A_t^\top & C_t^\top\end{bmatrix}+\begin{bmatrix}B_t\\ D_t\end{bmatrix}\begin{bmatrix}B_t^\top & D_t^\top\end{bmatrix}.\end{aligned}\tag{20}$$

Thus, we can apply Theorem 1 with $f\mapsto\bar f_t$, $\tilde f\mapsto\tilde f_t$ and $g\mapsto g_t$ and the least favorable density, i.e. the maximizer of (17), is

$$\tilde f_t(z_t|Y_{t-1})\sim\mathcal N(m_{z_t},\tilde K_{z_t})\tag{21}$$

where

$$\tilde K_{z_t}=\begin{bmatrix}\tilde K_{x_{t+1}} & K_{x_{t+1}y_t}\\ K_{y_tx_{t+1}} & K_{y_t}\end{bmatrix}$$

and

$$\begin{aligned}P_{t+1}&=K_{x_{t+1}}-K_{x_{t+1},y_t}K_{y_t}^{-1}K_{y_t,x_{t+1}}\\ V_{t+1}&=\tilde K_{x_{t+1}}-K_{x_{t+1},y_t}K_{y_t}^{-1}K_{y_t,x_{t+1}},\end{aligned}$$

and

$$\tilde{P}_{t+1} = \begin{cases} L_{P_{t+1}} \left(H_t^\top H_t - \theta_t (1-\tau) L_{P_{t+1}}^\top L_{P_{t+1}} \right)^{-\dagger \frac{1}{1-\tau}} L_{P_{t+1}}^\top, & 0 \le \tau < 1 \\ L_{P_{t+1}} \exp\left(\theta_t L_{P_{t+1}}^\top L_{P_{t+1}} \right) L_{P_{t+1}}^\top, & \tau = 1 \end{cases} \tag{22}$$

where $L_{P_{t+1}}$ is such that $P_{t+1} = L_{P_{t+1}} L_{P_{t+1}}^\top$, $H_t^\top H_t$ is the projection matrix with image equal to $\mathrm{Im}(P_{t+1})$ and $\theta_t < ((1-\tau)\sigma_{\max}(P_{t+1}))^{-1}$ is the unique solution to $\gamma_\tau(P_{t+1}, \theta_t) = c_t$ with γ_τ defined in (8). Moreover, the robust state estimator is

$$\hat{x}_{t+1} = A_t \hat{x}_t + G_t (y_t - C_t \hat{x}_t) \tag{23}$$

with $G_t = K_{x_{t+1}, y_t} K_{y_t}^{-1}$. By (20), we have

$$\begin{aligned} G_t &= (A_t \tilde{P}_t C_t^\top + B_t D_t^\top)(C_t \tilde{P}_t C_t^\top + D_t D_t^\top)^{-1} \\ P_{t+1} &= A_t \tilde{P}_t A_t^\top - G_t (C_t \tilde{P}_t C_t^\top + D_t D_t^\top) G_t^\top + B_t B_t^\top. \end{aligned} \tag{24}$$

Algorithm 1 presents the iterative scheme for the optimal robust estimator. While the latter shares its general framework with Algorithm 1 in Yi and Zorzi (2022), it differs in the distortion applied to P_{t+1} in Step 6, which depends on parameter τ; by varying τ, our algorithm generalizes the approach in Yi and Zorzi (2022) (for $\tau = 0$ it coincides with the specific instance of Yi & Zorzi, 2022) and allows a more flexible design of the model uncertainty. Algorithm 1 shares the iterative structure of the Kalman filter, except that P_t is perturbed through matrix $\tilde{P}_t$.

Remark 1. In the case the nominal model (12) is non-Gaussian, the proposed robust Kalman filter preserves the notion of optimality in the sense of being the best linear unbiased estimator based on second-order statistics within the considered ambiguity set.

Remark 2. The computational complexity of Algorithm 1 has been discussed in Remark 4 of Yi and Zorzi (2022) for the special case $\tau = 0$. The conclusion extends to the general case $0 \le \tau \le 1$, since both matrix inversion and the computation of matrix functions such as the matrix power and the matrix exponential have the same complexity, i.e. $O(n^3)$ where n is the dimension of the matrix. Thus, the computational complexity of Algorithm 1 is

$$O(n^3) + O\big(n(r_t+p)p\big) + O\big(p^2(r_t+p)\big) + O\big(n^2(r_t+p)\big) + O\left(\log_2 \left(\frac{\sigma_{\max}(P_{t+1})^{-1}}{\varepsilon} \right) n^3 \right)$$

where $\varepsilon > 0$ is the selected accuracy for solving the equation in Step 5, i.e. it is guaranteed that $|c_t - \gamma_\tau(P_{t+1}, \theta_t)| \le \varepsilon$, by means of the bisection method outlined in Zenere and Zorzi (2018).

The least favorable transition density $\tilde{\phi}_t^\circ(z_t|x_t)$ still needs to be characterized. Notice that

$$K_{z_t}^\dagger = \begin{bmatrix} I_n & 0 \\ -G_t^\top & I_p \end{bmatrix} \begin{bmatrix} P_{t+1}^\dagger & 0 \\ 0 & K_{y_t}^{-1} \end{bmatrix} \begin{bmatrix} I_n & -G_t \\ 0 & I_p \end{bmatrix}$$

$$\tilde{K}_{z_t}^\dagger = \begin{bmatrix} I_n & 0 \\ -G_t^\top & I_p \end{bmatrix} \begin{bmatrix} \tilde{P}_{t+1}^\dagger & 0 \\ 0 & K_{y_t}^{-1} \end{bmatrix} \begin{bmatrix} I_n & -G_t \\ 0 & I_p \end{bmatrix}$$

and thus

$$K_{z_t}^\dagger - \tilde{K}_{z_t}^\dagger = \begin{bmatrix} I_n \\ -G_t^\top \end{bmatrix} \Phi_t \begin{bmatrix} I_n & -G_t \end{bmatrix}$$

with $\Phi_t = P_{t+1}^\dagger - \tilde{P}_{t+1}^\dagger$ which is positive semi-definite and $\mathrm{Im}(\Phi_t) = \mathrm{Im}(H_t^\top)$. Then, we have that

$$\begin{aligned} &(z_t - m_{z_t})^\top (K_{z_t}^\dagger - \tilde{K}_{z_t}^\dagger)(z_t - m_{z_t}) \\ &\quad = \| x_{t+1} - (A_t \hat{x}_t + G_t (y_t - C_t \hat{x}_t)) \|_{\Phi_t}^2 = \| e_{t+1} \|_{\Phi_t}^2 \end{aligned}$$

Algorithm 1: Robust Kalman filter at time t.

Input : c_t, τ, $\hat{x}_t$, $\tilde{P}_t$, y_t
Output: $\hat{x}_{t+1}$, $\tilde{P}_{t+1}$

1 $G_t = (A_t \tilde{P}_t C_t^\top + B_t D_t^\top)(C_t \tilde{P}_t C_t^\top + D_t D_t^\top)^{-1}$
2 $\hat{x}_{t+1} = A_t \hat{x}_t + G_t (y_t - C_t \hat{x}_t)$
3 $P_{t+1} = A_t \tilde{P}_t A_t^\top - G_t (C_t \tilde{P}_t C_t^\top + D_t D_t^\top) G_t^\top + B_t B_t^\top$
4 Take $H_t^\top H_t$ projection matrix s.t. $\mathrm{Im}(H_t^\top H_t) = \mathrm{Im}(P_{t+1})$
5 Find θ_t such that $c_t = \gamma_\tau(P_{t+1}, \theta_t)$
6

$$\tilde{P}_{t+1} = \begin{cases} L_{P_{t+1}} \left(H_t^\top H_t - \theta_t (1-\tau) L_{P_{t+1}}^\top L_{P_{t+1}} \right)^{\dagger \frac{1}{1-\tau}} L_{P_{t+1}}^\top, & 0 \le \tau < 1 \\ L_{P_{t+1}} \exp\left(\theta_t L_{P_{t+1}}^\top L_{P_{t+1}} \right) L_{P_{t+1}}^\top, & \tau = 1 \end{cases}$$

where $e_t = x_t - \hat{x}_t$ denotes the prediction error corresponding to the robust filter derived above. Thus, in view of (21) and (19), we have that the least favorable density is

$$\begin{aligned} \tilde{f}_t(z_t|Y_{t-1}) &\sim \exp\left(\frac{1}{2} \| e_{t+1} \|_{\Phi_t}^2 \right) \bar{f}_t(z_t|Y_{t-1}) \\ &\sim \exp\left(\frac{1}{2} \| e_{t+1} \|_{\Phi_t}^2 \right) \int \phi_t(z_t|x_t) \tilde{f}_t(x_t|Y_{t-1}) \mathrm{d}x_t \end{aligned}$$

In view of (15), the maximizer of (17) is

$$\tilde{\phi}_t(z_t|x_t) = \frac{1}{M_t(\Phi_t)} \exp\left(\frac{1}{2} \| e_{t+1} \|_{\Phi_t}^2 \right) \phi_t(z_t|x_t) \tag{25}$$

where the normalizing constant $M_t(\Phi_t)$ is such that (18) holds. Notice that, using the τ-divergence with $\tau > 0$ introduces a distortion which is a non-radial function of e_{t+1} along directions in $\mathrm{Im}(H_t^\top)$, in contrast to the case $\tau = 0$.

4. Least favorable model

To simulate and evaluate the filter performance, particularly when selecting parameters c_t and τ, it is crucial to characterize the least favorable model corresponding to (25). This, in turn, allows one to assess whether the prescribed uncertainty is realistic for the considered application. In view of (12), z_t belongs to the affine space

$$\mathbb{S} = \{ [\, A_t^\top \; C_t^\top \,]^\top x_t + \bar{\imath} \bar{w}_t, \; \bar{\imath} \bar{w}_t \in \mathrm{Im}(\Gamma_t) \}$$

and there is a one to one correspondence between z_t and v_t, given x_t, through the isomorphism

$$\mathbb{S} \to \mathbb{R}^{r_t+p}$$

$$z_t \mapsto v_t = (\Gamma_t^\top \Gamma_t)^{-1} \Gamma_t^\top \left(z_t - \begin{bmatrix} A_t \\ C_t \end{bmatrix} x_t \right).$$

Hence, the least favorable model can be characterized through model (12) in which the distortion has been moved now in v_t. Applying the same arguments used in (Yi & Zorzi, 2022, Section V), we have that the least favorable density of v_t depends on e_t, i.e. the maximizer has the opportunity to change retroactively the least favorable density of x_t, and is distributed as $\tilde{\psi}_t(v_t|e_t) \sim \mathcal{N}(F_t e_t, \tilde{K}_{v_t})$ where

$$\tilde{K}_{v_t} = (I_{r_t+p} - (B_t - G_t D_t)^\top (\Omega_{t+1}^\dagger + \Phi_t)(B_t - G_t D_t))^{-1}$$

and

$$F_t = \tilde{K}_{v_t} (B_t - G_t D_t)^\top (\Omega_{t+1}^\dagger + \Phi_t)(A_t - G_t C_t).$$

Matrix $\Omega_{t+1}^\dagger$ is computed from the backward recursion

$$\Omega_t^\dagger = (A_t - G_t C_t)^\top (\Omega_{t+1}^\dagger + \Phi_t)(A_t - G_t C_t) + F_t^\top \tilde{K}_{v_t}^{-1} F_t$$

with final condition $\Omega^{\dagger}_{T+1} = 0$ and T is the simulation horizon. Moreover, the least favorable model admits a state space representation with matrices

$$\tilde{A}_t = \begin{bmatrix} A_t & B_t F_t \\ 0 & A_t - G_t C_t + (B_t - G_t D_t)F_t \end{bmatrix},$$

$$\tilde{B}_t = \begin{bmatrix} B_t \\ B_t - G_t D_t \end{bmatrix} J_t,\ \tilde{C}_t = \begin{bmatrix} C_t & D_t F_t \end{bmatrix},\ \tilde{D}_t = D_t J_t,$$

J_t is such that $\tilde{K}_{v_t} = J_t J_t^{\top}$ and the input is white Gaussian noise with unit variance. To construct the least favorable model, we first compute the gain G_t via a forward sweep of the robust filter (23)-(24), followed by generating the matrices Ω_t through a backward sweep. Finally, the performance of an estimator of the form $\hat{x}'_{t+1} = A_t \hat{x}'_t + G'_t(y_t - C_t \hat{x}'_t)$ applied to the least favorable model is obtained by the Lyapunov equation

$$\begin{aligned}\Pi_{t+1} =& \left(\tilde{A}_t - \begin{bmatrix} G'_t \\ 0 \end{bmatrix} \tilde{C}_t\right) \Pi_t \left(\tilde{A}_t - \begin{bmatrix} G'_t \\ 0 \end{bmatrix} \tilde{C}_t\right)^{\top} \\ &+ \left(\tilde{B}_t - \begin{bmatrix} G'_t \\ 0 \end{bmatrix} \tilde{D}_t\right)\left(\tilde{B}_t - \begin{bmatrix} G'_t \\ 0 \end{bmatrix} \tilde{D}_t\right)^{\top}\end{aligned}$$

where

$$\Pi_t = \mathbb{E}\left[\begin{bmatrix} e'_t \\ e_t \end{bmatrix}\begin{bmatrix} (e'_t)^{\top} & e_t^{\top} \end{bmatrix}\right],$$

$e'_t = x_t - \hat{x}'_t$ and e_t is the estimation error of the optimal filter (23). Notice that, the time-varying matrices computed in the backward iteration are determined solely by the nominal model and do not depend on the observations y_t. Therefore, they can be computed offline before observations become available, allowing the least favorable model to be applied online.

5. Convergence analysis

We study the convergence of the robust Kalman filter proposed in Section 3 under the assumption that the nominal model has constant parameters, that is $A_t = A$, $B_t = B$, $C_t = C$ and $D_t = D$, and the tolerance is constant, i.e. $c_t = c$. We show that, if c is sufficiently small, then the convergence of the filter is guaranteed. Without loss of generality, we assume that $BD^{\top} = 0$. Otherwise, a filter of the type described in Algorithm 1 can be equivalently rewritten with

$$\tilde{A} = A - BD^{\top}(DD^{\top})^{-1}C, \quad \tilde{C} = C, \quad \tilde{D} = D,$$

$\tilde{B}$ such that $\tilde{B}\tilde{B}^{\top} = B(I - D^{\top}(DD^{\top})^{-1}D)B^{\top}$,

so that $\tilde{B}\tilde{D}^{\top} = 0$, see Lancaster and Rodman (1995), Zorzi (2017a). The robust filter converges if and only if the iteration defined on $\overline{\mathcal{Q}^n_+}$

$$P_{t+1} = r_c(P_t), \quad P_0 \in \overline{\mathcal{Q}^n_+} \tag{26}$$

converges. The mapping r_c is defined as follows

$$r_c(P_t) = A\tilde{P}_t A^{\top} - A\tilde{P}_t C^{\top}(C\tilde{P}_t C^{\top} + R)^{-1} C\tilde{P}_t A^{\top} + Q,$$

where $R := DD^{\top}$, $Q := BB^{\top}$, $\tilde{P}_{t+1}$ is defined in (22) and θ_t satisfies the condition $\gamma_\tau(P_{t+1}, \theta_t) = c$.

Proposition 1. *Let $\bar{P}_t$, $t \geq 0$, be the sequence generated by the Riccati equation*

$$\bar{P}_{t+1} = A\bar{P}_t A^{\top} - A\bar{P}_t C^{\top}(C\bar{P}_t C^{\top} + R)^{-1} C\bar{P}_t A^{\top} + Q$$

with $\bar{P}_0 = P_0$. Then, the sequence P_t, $t \geq 0$, generated by (26) is such that $\mathrm{Im}(P_t) = \mathrm{Im}(\bar{P}_t)$.

Proof. The proof is similar to the one of (Yi & Zorzi, 2022, Proposition 1) and it exploits the fact that $\mathrm{Im}(\tilde{P}_t) = \mathrm{Im}(P_t)$. □

If the pair (A, B) is stabilizable and (A, C) is detectable, $\bar{P}_t$ converges to a unique solution, denoted $\bar{P}_\infty$, for any initial condition $\bar{P}_0$, Lancaster and Rodman (1995). Let U and V be matrices whose columns form an orthonormal basis for $\mathrm{Im}(\bar{P}_\infty)$ and its orthogonal complement in $\mathbb{R}^n$, respectively, and assume U and V are fixed. In view of Proposition 1, there exists $\bar{t} \in \mathbb{N}$ such that

$$P^R_t := U^{\top} P_t U \in \mathcal{Q}^r_+, \quad \forall\, t \geq \bar{t}. \tag{27}$$

In what follows we show, taking c sufficiently small, that P^R_t converges and $V^{\top} P_t V \to 0$ for any initial condition and thus P_t converges.

Proposition 2. *Consider the sequence generated by (26) with $\mathrm{Im}(P_0) \subseteq \mathrm{Im}(U)$. Assume (A, B) is stabilizable and the reachable subsystem is observable. Then, there exists $c_{MAX} > 0$ such that for any $0 < c < c_{MAX}$, the sequence in (27) converges to a unique point P^R_∞. Accordingly, the sequence in (26) converges to $P_\infty := U P^R_\infty U^{\top}$ for $0 < c < c_{MAX}$.*

Proof. Using the same arguments used in the proof of (Yi & Zorzi, 2022, Lemma 2), we have that $\mathrm{Im}(P_t) \subseteq \mathrm{Im}(U)$, $t > 0$, and the subsystem $(U^{\top}AU, U^{\top}B, CU, D)$ is reachable. Hence, we have that $P_t = U P^R_t U^{\top}$, $t \geq 0$, and by (27) we have that $\mathrm{Im}(P_t) = \mathrm{Im}(U)$ for $t \geq \bar{t}$. Thus, the iteration (26) for $t \geq \bar{t}$ is equivalent to the "reduced" iteration

$$P^R_{t+1} = U^{\top} r_c(P_t) U = U^{\top} r_c(U P^R_t U^{\top}) U = r^R_c(P^R_t) \tag{28}$$

where

$$r^R_c(P^R_t) := A_R((\tilde{P}^R_t)^{-1} + C_R^{\top} R^{-1} C_R)^{-1} A_R^{\top} + Q_R$$

$A_R := U^{\top}AU$, $Q_R := U^{\top}BB^{\top}U$, $C_R := CU$, $\tilde{P}^R_t := U^{\top}\tilde{P}_t U \in \mathcal{Q}^r_+$ and we exploited the fact that $\mathrm{Im}(H_t^{\top}) = \mathrm{Im}(P_t) = \mathrm{Im}(U)$, for $t \geq \bar{t}$. Consider the subsequence $P^{Rd}_k := P^R_{kN}$ with $k \geq \lceil \bar{t}/N \rceil$. The corresponding recursion is $P^{Rd}_{k+1} = r^{Rd}_c(P^{Rd}_k)$ where $r^{Rd}_c(\cdot)$ is the N-fold composition of $r^R_c(\cdot)$. Since the system $(A_R, Q_R^{1/2}, C_R, D)$ is reachable and observable, by (Zorzi, 2017a, Theorem 4.4), there exists $c_{\mathrm{MAX}} > 0$ such that the iteration in (28) converges to a unique point $P^R_\infty \in \mathcal{Q}^I_+$ for any $P_0 \in \overline{\mathcal{Q}^n_+}$ such that $\mathrm{Im}(P_0) \subseteq \mathrm{Im}(U)$. □

Next, we address the general case, i.e. when $\mathrm{Im}(P_0) \not\subseteq \mathrm{Im}(U)$. In this case the image of P_t is not necessarily contained in $\mathrm{Im}(U)$.

Lemma 2. *Consider the sequence generated by (26) with $P_0 \in \overline{\mathcal{Q}^n_+}$. Assume that (A, B) is stabilizable and (A, C) is detectable. Then, there always exists $c_{MAX} > 0$ such that if $0 < c < c_{MAX}$ then $V^{\top} P_t V \to 0$.*

Proof. We prove the claim for $0 < \tau < 1$, as the cases $\tau = 0$ and $\tau = 1$ follow similarly. Considering the reachability standard form as done in the proof of (Yi & Zorzi, 2022, Lemma 4) it is possible to obtain $\check{P}_{22,t+1} \leq A_{22}\check{\tilde{P}}_{22,t}A_{22}^{\top}$ where A_{22} is the stable matrix of the non-reachable subspace and $\check{P}_{22,t}$, $\check{\tilde{P}}_{22,t}$ are the nominal and least favorable covariance matrices of the prediction error in the non-reachable subspace. Since $\tilde{P}_t$, P_t have the same eigenvectors corresponding to the nonnull eigenvalues, then condition $\gamma_\tau(P_t, \theta_{t-1}) = c$ is equivalent to

$$\sum_i -\frac{1}{\tau(1-\tau)}\left(\frac{\sigma_i(\tilde{P}_t)}{\sigma_i(P_t)}\right)^{\tau} + \frac{1}{1-\tau}\frac{\sigma_i(\tilde{P}_t)}{\sigma_i(P_t)} + \frac{1}{\tau} = c \tag{29}$$

where $\sigma_i(P_t)$ and $\sigma_i(\tilde{P}_t)$ denote the i-th largest nonnull eigenvalue of P_t and $\tilde{P}_t$, respectively. Since the terms on the left hand side of (29) are nonnegative, it follows that

$$-\frac{1}{\tau(1-\tau)}\left(\frac{\sigma_i(\tilde{P}_t)}{\sigma_i(P_t)}\right)^{\tau} + \frac{1}{1-\tau}\frac{\sigma_i(\tilde{P}_t)}{\sigma_i(P_t)} + \frac{1}{\tau} \leq c. \tag{30}$$

In view of (30) we have $\tilde{P}_t \leq \rho P_t$, and thus $\check{\tilde{P}}_{22,t} \leq \rho \check{P}_{22,t}$, where $\rho > 1$ is the unique solution to the equation

$$-\frac{1}{\tau(1-\tau)}\rho^{\tau} + \frac{1}{1-\tau}\rho + \frac{1}{\tau} = c. \tag{31}$$

We conclude that $\check{P}_{22,t+1} \leq \rho A_{22}\check{P}_{22,t}A_{22}^{\top}$. Accordingly, if $\sqrt{\rho}A_{22}$ is stable, then $\check{P}_{22,t} \to 0$ and $\check{P}_{22,t}$ corresponds to $V^{\top}P_t V$ in the original coordinates. The existence of $c_{\mathrm{MAX}} > 0$ such that $\sqrt{\rho}A_{22}$ is stable follows from the fact that the left-hand side of (31) is monotonically increasing for $\rho > 1$ and approaches zero as $\rho \to 1$. □

Theorem 2. *Consider the sequence generated by (26) with $P_0 \in \overline{\mathcal{Q}}_+^n$. Assume that (A, B) is stabilizable and the reachable subsystem is also observable. Then, there always exists $c_{MAX} > 0$ such that if $0 < c < c_{MAX}$ then the sequence in (26) converges to $P_\infty = U P_\infty^R U^\top$ for any initial condition $P_0 \in \overline{\mathcal{Q}}_+^n$.*

Proof. Consider the fixed point P_∞^R, P_k^{Rd} and $P_k^d := P_{kN}$ with $k \geq \lceil \bar{t}/N \rceil$, from Proposition 2 where $N \geq n$ and $\bar{t}$ is chosen so that (27) holds. Consequently, both P_∞^R and P_k^{Rd} belong to the metric space $(\mathcal{Q}_+^l, \delta)$ where δ denotes the Thompson part metric, Gaubert and Qu (2014). Then, we have

$$P_{k+1}^d = r_c^d(P_k^d), \qquad P_{k+1}^{Rd} = r_c^{Rd}(P_k^d),$$

where $r_c^{Rd}(\cdot)$ has been defined in Proposition 2 and $r_c^d(\cdot)$ denotes the N-fold compositions of $r_c(\cdot)$. Since $r_c(\cdot)$ is a continuous mapping on $\overline{\mathcal{Q}}_+^n$, its N-fold composition $r_c^d(\cdot)$ is also continuous on the same set. By Lemma 2, $V^\top P_t V \to 0$ and thus $V^\top P_k^d V \to 0$. Hence, $r_c^d(P_k^d) = r_c^d(UU^\top P_k^d UU^\top) + M_k$, where $M_k \to 0$. Thus, we have

$$\begin{aligned} P_{k+1}^{Rd} &= U^\top r_c^d(P_k^d)U = U^\top r_c^d(UU^\top P_k^d UU^\top)U + N_k \\ &= r_c^{Rd}(P_k^{Rd}) + N_k, \end{aligned}$$

where $N_k = U^\top M_k U \in \mathcal{Q}^l$ and $\|N_k\| \to 0$ as $k \to \infty$. Since $r_c^{Rd}(\cdot)$ is strict contraction in $(\mathcal{Q}_+^l, \delta)$ with contraction coefficient $\eta_N < 1$, Zorzi (2017a), we have that

$$\begin{aligned} &\delta(P_{k+1}^{Rd}, P_\infty^R) = \delta(r_c^{Rd}(P_k^{Rd}) + N_k, P_\infty^R) \\ &= \delta(r_c^{Rd}(P_k^{Rd}), P_\infty^R) + \varepsilon_k = \delta(r_c^{Rd}(P_k^{Rd}), r_c^{Rd}(P_\infty^R)) + \varepsilon_k \\ &\leq \eta_N\, \delta(P_k^{Rd}, P_\infty^R) + \varepsilon_k, \end{aligned}$$

where $\varepsilon_k \to 0$ as $k \to \infty$ because the Thompson part metric $\delta(\cdot,\cdot)$ is continuous with respect to its arguments. It follows that

$$\delta(P_{k+1}^{Rd}, P_\infty^R) \leq \eta_N^{k+1}\delta(P_0^{Rd}, P_\infty^R) + \sum_{l=0}^{k} \eta_N^{k-l}\varepsilon_l$$

Since $\eta_N < 1$, $\delta(P_{k+1}^{Rd}, P_\infty^R) \to 0$ and thus $P_k^{Rd} \to P_\infty^R$ as $k \to \infty$. The same argument applies to subsequences of the form $P_k^{Rd} := P_{kN+l}^R$, with $l = 1, \ldots, N-1$. Consequently, $P_t^R \to P_\infty^R$ as $t \to \infty$. □

6. Numerical experiments

The numerical experiments presented in this section are intended to provide some insight on the selection of the parameter τ. Consider the nominal state space model (12) with:

$$A = \begin{bmatrix} 2 & 0.1 & 0.1 \\ 0 & 0.8962 & -0.3712 \\ 0 & 0.3712 & 0.8962 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix},$$
$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{32}$$

The aforementioned model is not reachable; however, the pairs (A, B) and (A, C) are stabilizable and observable, respectively. Accordingly, the Riccati iteration is not defined on the cone of the positive definite matrices and thus the RKF proposed in Zorzi (2016) cannot be applied. We set the covariance matrix of the initial state equal to $\tilde{P}_0 = 6 \cdot I_3$. We consider the proposed robust Kalman filter (denoted by RKF) with $c = 5 \cdot 10^{-1}$ and with three different values for τ: $\tau = 0$ (corresponding to RKF proposed in Yi & Zorzi, 2022), $\tau = 0.5$ and $\tau = 1$.

Fig. 1 shows the risk sensitivity parameter θ_t of RKF for the different values of τ. In all the cases θ_t converges to a constant value as time increases. Moreover, the larger τ is, the larger the steady-state value of θ_t becomes. It is worth noting that this does not mean that the RKF with $\tau = 1$ is more risk-aware than the other filters, but rather that the way uncertainty is quantified in the ambiguity set changes with τ. Next, we also consider the standard Kalman filter (denoted by KF).

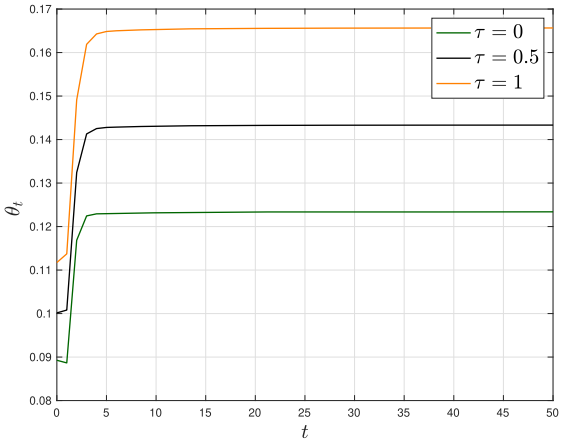


Fig. 1. Risk sensitivity parameter θ_t for RKF.

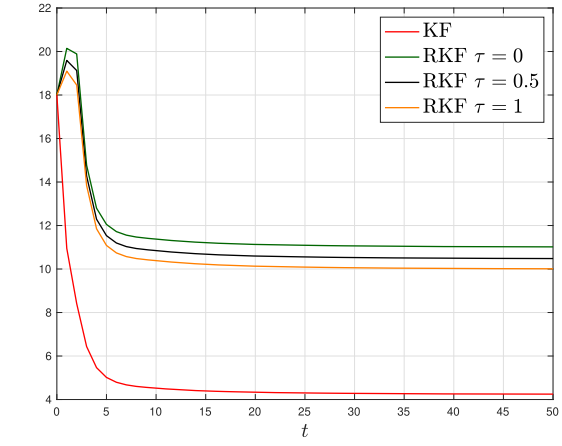


Fig. 2. Trace of $\tilde{P}_t$ for RKF versus the trace of P_t for KF.

In Fig. 2, we compare the trace of $\tilde{P}_t$ for RKF and the trace of P_t of KF[1] We can notice that $\mathrm{tr}(\tilde{P}_t) > \mathrm{tr}(P_t)$ meaning that all RKF propagate more uncertainty than KF in the state estimation. Moreover, $\mathrm{tr}(\tilde{P}_t)$ converges to a constant value as time increases for all the selected values of τ, confirming the convergence of RKF if tolerance c is taken sufficiently small.

Let $\mathcal{M}(c, \tau)$ denote the least favorable model corresponding to the maximizer of (17) and with simulation horizon $T = 200$. Clearly, if the actual model is $\mathcal{M}(c, \tau)$, then the best performance is attained by RKF with ambiguity set whose shape is defined by τ and radius by c. Next, we test the performance of the robust filters with $c = 5 \cdot 10^{-1}$ and $\tau \in \{0, 0.5, 1\}$ in the case where the actual model is not the least favorable model for which they were designed. More precisely, we consider as actual model $\mathcal{M}(c_{\mathrm{LF}}, \tau_{\mathrm{LF}})$ with c_{LF} different from c.

Fig. 3 shows the scalar variance of the prediction error for the KF and the robust filters in the case where the actual model is $\mathcal{M}(c_{\mathrm{LF}}, \tau_{\mathrm{LF}})$ with $c_{\mathrm{LF}} = 6 \cdot 10^{-1}$ and $\tau_{\mathrm{LF}} = 1$. Fig. 4 corresponds to the case with $c_{\mathrm{LF}} = 6 \cdot 10^{-1}$ and $\tau_{\mathrm{LF}} = 0.5$, and Fig. 5 to the case with $c_{\mathrm{LF}} = 6 \cdot 10^{-1}$ and $\tau_{\mathrm{LF}} = 0$.

In all cases, the robust estimators achieve the same steady-state performance and outperform the Kalman filter, even though the radius of their ambiguity sets is smaller than that of the actual one. The differ-

[1] Notice that, KF does coincide with RKF with $c = 0$ (i.e. the ambiguity set only contains the nominal model) and in that case $\tilde{P}_t = P_t$.

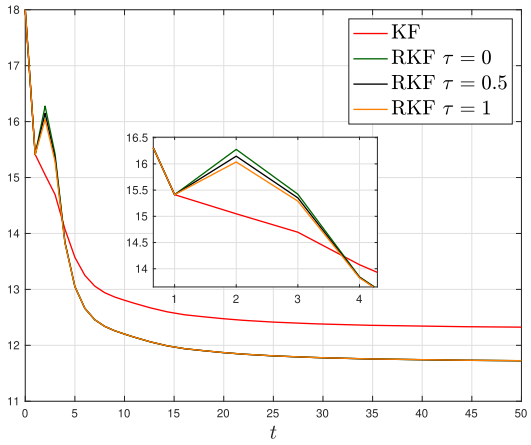


Fig. 3. Scalar variance of the prediction error for KF and RKF with $c = 5 \cdot 10^{-1}$ and $\tau \in \{0, 0.5, 1\}$ in the case the actual model is $\mathcal{M}(c_{\text{LF}}, \tau_{\text{LF}})$ with $c_{\text{LF}} = 6 \cdot 10^{-1}$ and $\tau_{\text{LF}} = 1$.

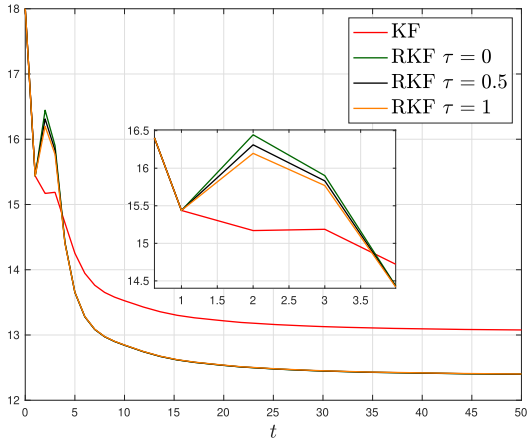


Fig. 4. Scalar variance of the prediction error for KF and RKF with $c = 5 \cdot 10^{-1}$ and $\tau \in \{0, 0.5, 1\}$ in the case that the actual model is $\mathcal{M}(c_{\text{LF}}, \tau_{\text{LF}})$ with $c_{\text{LF}} = 6 \cdot 10^{-1}$ and $\tau_{\text{LF}} = 0.5$.

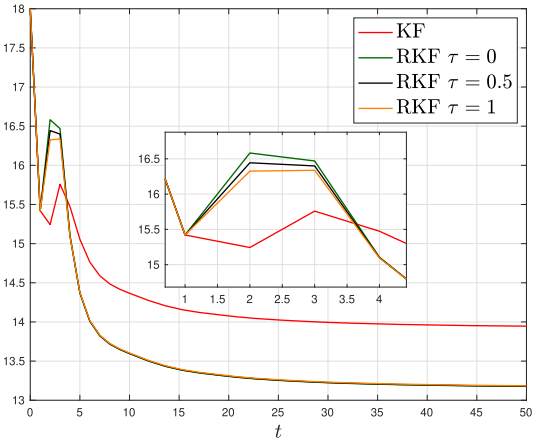


Fig. 5. Scalar variance of the prediction error for KF and RKF with $c = 5 \cdot 10^{-1}$ and $\tau \in \{0, 0.5, 1\}$ in the case the actual model is $\mathcal{M}(c_{\text{LF}}, \tau_{\text{LF}})$ with $c_{\text{LF}} = 6 \cdot 10^{-1}$ and $\tau_{\text{LF}} = 0$.

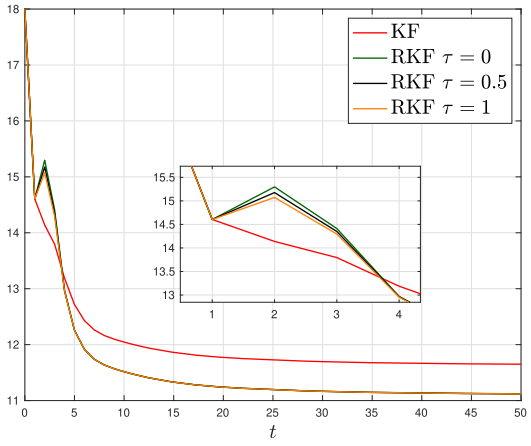


Fig. 6. Scalar variance of the prediction error for KF and RKF with $c = 5 \cdot 10^{-1}$ and $\tau \in \{0, 0.5, 1\}$ in the case the actual model is $\mathcal{M}(c_{\text{LF}}, \tau_{\text{LF}})$ with $c_{\text{LF}} = 4 \cdot 10^{-1}$ and $\tau_{\text{LF}} = 0$.

ence among the robust Kalman filters appears in the transient phase, where RKF with $c = 5 \cdot 10^{-1}$ and $\tau = 1$ exhibits the best performance, even when the actual ambiguity set has a different shape (e.g., $\tau_{LF} = 0$; see Fig. 5). We also considered the setup in which the ambiguity set of RKF is with a larger radius than the actual one, and we obtained similar results. For instance, Fig. 6 shows the scalar variance of the prediction error for KF and the robust filters in the case where the actual model is $\mathcal{M}(c_{\text{LF}}, \tau_{\text{LF}})$, with $c_{\text{LF}} = 4 \cdot 10^{-1}$ and $\tau_{\text{LF}} = 0$.

In all previous experiments, the tolerance has been assumed to be known. However, in practice, it is not. A possible solution consists in estimating the tolerance parameter c directly from the available data. Given a collection of observations and a predefined set of admissible values for c, the selected tolerance is the one associated with the robust predictor that yields the lowest empirical variance of the prediction residuals (Yi et al., 2025).

7. Conclusions

In this paper, we proposed a family of robust Kalman filters corresponding to the ambiguity set which is a ball formed by placing an upper bound on the τ-divergence. The shape of the ambiguity set is defined by parameter τ and its radius c. Unlike the former solution proposed in Zorzi (2016), these filters are well defined even when the covariance matrix is positive semi-definite but not invertible. We have derived the corresponding least favorable model and analyzed the convergence of the corresponding Riccati iteration in the case the nominal model has constant parameters. Finally, the numerical results showed that: (i) even though the actual ambiguity set was not precisely known, the robust filters still outperformed the standard Kalman filter; and (ii) introducing the τ-divergence to define the ambiguity set provided a more flexible filtering approach, with the RKF for $\tau = 1$ appearing to be more robust during the transient phase and outperforming RKF for $\tau = 0$ (i.e. the one proposed in Yi and Zorzi (2022)).

CRediT authorship contribution statement

Mattia Zorzi: Methodology, Supervision; **Shenglun Yi:** Methodology, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Başar, T., & Bernhard, P. (2008). H-infinity optimal control and related minimax design problems: A dynamic game approach. Springer Science & Business Media.

Bocquet, M., Gurumoorthy, K. S., Apte, A., Carrassi, A., Grudzien, C., & Jones, C. K. (2017). Degenerate Kalman filter error covariances and their convergence onto the unstable subspace. *The SIAM/ASA Journal on Uncertainty Quantification*, *5*(1), 304–333.

Bonnabel, S., & Sepulchre, R. (2013). The geometry of low-rank Kalman filters. In *Matrix information geometry* (pp. 53–68). Springer.

Gaubert, S., & Qu, Z. (2014). The contraction rate in Thompson's part metric of order-preserving flows on a cone - application to generalized Riccati equations. *Journal of Differential Equations*, *256*(8), 2902–2948.

Hansen, L., & Sargent, T. (2008). Robustness. Princeton, NJ: Princeton University Press.

Kailath, T., Sayed, A., & Hassibi, B. (2000). Linear estimation. Upper Saddle River, NJ: Prentice Hall.

Kim, S., Deshpande, V., & Bhattacharya, R. (2020). Robust Kalman filtering with probabilistic uncertainty in system parameters. *The IEEE Control Systems Letters*, *5*(1), 295–300.

Lancaster, P., & Rodman, L. (1995). Algebraic Riccati equations. Clarendon Press.

Levy, B., & Nikoukhah, R. (2004). Robust least-squares estimation with a relative entropy constraint. *The IEEE Transactions on Information Theory*, *50*(1), 89–104.

Levy, B., & Nikoukhah, R. (2013). Robust state-space filtering under incremental model perturbations subject to a relative entropy tolerance. *IEEE Transactions on Automatic Control*, *58*, 682–695.

Nguyen, V. A., Shafieezadeh-Abadeh, S., Kuhn, D., & Esfahani, P. M. (2023). Bridging Bayesian and minimax mean square error estimation via Wasserstein distributionally robust optimization. *Mathematical Methods of Operations Research*, *48*(1), 1–37.

Song, E., Xu, J., & Zhu, Y. (2014). Optimal distributed Kalman filtering fusion with singular covariances of filtering errors and measurement noises. *IEEE Transactions on Automatic Control*, *59*(5), 1271–1282. https://doi.org/10.1109/TAC.2014.2308451

Yi, S., Jin, X., Wang, Z., Liu, Z., & Zorzi, M. (2025). Data-driven robust UAV position estimation in GPS signal-challenged environment. In *ACC* (pp. 491–496).

Yi, S., & Zorzi, M. (2022). Robust Kalman filtering under model uncertainty: The case of degenerate densities. *IEEE Transactions on Automatic Control*, *67*(7), 3458–3471.

Yoon, M., Ugrinovskii, V., & Petersen, I. (2004). Robust finite horizon minimax filtering for discrete-time stochastic uncertain systems. *Systems & Control Letters*, *52*(2), 99–112.

Zenere, A., & Zorzi, M. (2018). On the coupling of model predictive control and robust Kalman filtering. *IET Control Theory & Applications*, *12*(13), 1873–1881.

Zorzi, M. (2016). Robust Kalman filtering under model perturbations. *IEEE Transactions on Automatic Control*, *62*(6), 2902–2907.

Zorzi, M. (2017a). Convergence analysis of a family of robust Kalman filters based on the contraction principle. *SIAM Journal on Control and Optimization*, *55*(5), 3116–3131.

Zorzi, M. (2017b). On the robustness of the Bayes and Wiener estimators under model uncertainty. *Automatica*, *83*, 133–140.

Zorzi, M., & Yi, S. (2024). On the convergence of degenerate risk sensitive filters. *Systems & Control Letters*, *185*, 105732.